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GaPSE.jl: full-relativistic two-point statistics of galaxy clustering

Matteo Foglieni [1] **and Mattia Pantiri** [2]

1 Leibniz Supercomputing Centre (LRZ), Boltzmannstr. 1, 85748 Garching bei Muenchen, Germany **2** Leiden Institute of Physics (LION), Niels Bohrweg 2, 2333 CA Leiden, The Netherlands

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Software

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Summary

`GaPSE.jl` (Galaxy Power Spectrum Estimator) is an open-source Julia package for computing the two-point statistics of relativistic effects in galaxy clustering at first order in perturbation theory. It computes Two-Point Correlation Functions of relativistic Galaxy Number Counts and Luminosity Distance perturbations, their cross-correlations, and the corresponding power spectra multipoles, with or without a survey window function. The package allows users to compute individual contributions from the physical effects entering these observables, including Doppler effect, lensing, and gravitational potentials. This decomposition makes it possible to study the relative importance of each relativistic correction individually and their potential degeneracies with signals from local Primordial Non-Gaussianity, across different redshift bins, multipoles, and survey geometries.

Statement of need

The current standard paradigm for the origin of structure formation in the Universe is the inflation, i.e. an exponentially rapid growth of the Universe scale factor a of about 28 orders of magnitude. Almost all inflationary models predict a flat universe and a nearly scale-invariant spectrum of initial fluctuations. Consequently, identifying observables that can discriminate between these models is of utmost importance. Local Primordial Non-Gaussianities (PNG) are one of the most promising variables of this kind (Achúcarro et al., 2022).

The standard parameterization of the local PNG is the so-called scale-independent f_{NL} parameterization, which would imprint a scale-dependent bias at low wavenumbers on the observable Power Spectrum (Dalal et al., 2008; Slosar et al., 2008). However, on those scales, the pure Newtonian and plane-parallel approximations that underpin most clustering analyses cease to be adequate, because the separation between a pair of galaxies becomes comparable to the distance from us to them. Consequently, a full General Relativistic (GR) approach must be employed.

In such analysis, the GR effects manifest themselves in two groups of observables, that can be derived analytically (Castorina & Di Dio, 2022): the Luminosity Distance (LD) fluctuations and the Galaxy Number Counts (GNC). Due to the presence of the gravitational potential in many GR effects, some of these terms can mimic a PNG signal and bias its measurement. For this reason, we need a deeper understanding of these GR effects, especially for future surveys that aim to map the Universe on the largest scales, like Euclid (Laureijs et al., 2011). This is why we developed `GaPSE.jl`.

`GaPSE.jl` (Galaxy Power Spectrum Estimator) is an open-source code, written in the Julia programming language, that computes the two-point statistics of these GR effects at first order in perturbation theory. This enables users to assess the importance of relativistic corrections,

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compare them with the PNG signal, and investigate how survey geometry affects the resulting observables.

State of the field

Existing publicly available codes address related but distinct observables. Boltzmann codes such as `CAMB` (Challinor & Lewis, 2011; Lewis, 2026) and `CLASS` (Blas et al., 2011; Dio et al., 2013) calculate angular power spectra of observed number counts, $C_\ell(z_i, z_j)$. This representation is exact, but these angular spectra are not directly the Fourier-space Power Spectrum multipoles commonly used in galaxy clustering analyses, and the translation from one to the other is non-trivial. `GaPSE.jl` implements instead the Fourier-space formalism of Castorina & Di Dio (2022), in which the observed Power Spectrum multipoles are obtained from the Yamamoto estimator, using the endpoint line-of-sight convention. Consequently, its outputs can be compared with the measured multipoles obtained from spectroscopic surveys directly, without an intermediate conversion from angular spectra.

`COFFE` (COrrelation Function Full-sky Estimator) provides a complementary approach, calculating full-sky and flat-sky galaxy two-point correlation functions and their multipoles, including relativistic and wide-angle effects (Tansella et al., 2018). It also supports covariance calculations for relevant correlation-function estimators. These capabilities make `COFFE` an important existing alternative for analyses centered on galaxy number counts in configuration space. `GaPSE.jl` differs in three respects: it computes also the Power Spectra multipoles (including an arbitrary survey window), rather than configuration-space correlation functions only; it extends the analysis to Luminosity Distance perturbations and their cross correlations with Galaxy Number Counts; and it includes a model for the local PNG contribution. In addition, a single function call in `GaPSE.jl` returns every auto- and cross-term separately together with their sum. These capabilities make `GaPSE.jl` complementary to the currently existing tools and motivate its development as a separate package.

Software design

The design of `GaPSE.jl` separates cosmological inputs from the calculation of the observables. Users provide two input files: the linear matter power spectrum at redshift $z = 0$ and the background quantities of the chosen cosmology. Example files for a standard cosmology are provided in the `data` directory; users can generate equivalent files for their own cosmological models using `CLASS` (Blas et al., 2011). These inputs are loaded into a `Cosmology` Julia `struct`, which stores the cosmological parameters and the interpolating splines required by subsequent calculations. This allows multiple correlation functions and power-spectrum multipoles to be evaluated using the same cosmological setup.

Once a `Cosmology` object has been initialized, users can calculate the supported Two-Point Correlation Function (TPCF) multipoles for Galaxy Number Counts (GNC), Luminosity Distance (LD) perturbations, their cross-correlations, and the local Primordial Non-Gaussianity (PNG) contribution. Doppler and matter correlation functions in plane-parallel approximation are also available for comparison. Individual physical contributions are computed separately, rather than only as a total observable, allowing users to investigate their relative importance in the given `Cosmology`.

Survey geometry is incorporated through the `use_windows` boolean option. The package provides a toy-model window function with azimuthal symmetry for controlled tests and comparisons. For real applications, users can supply an arbitrary window function as a tabulated map.

The computational core of the package is the evaluation of the integrals for the TPCFs, which can involve, depending on the term considered, 1, 2 or 3 nested integrals. The four contributions

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involving three nested integrals (Lensing auto-correlation, Integrated Gravitational Potential auto-correlation, and their 2 cross-correlations) are by far the most computationally expensive contributions. The following profiling flame graph obtained with Intel VTune illustrates the computational workload involved in evaluating all 25 GNC TPCFs, and the predominance of the aforementioned 4 terms:

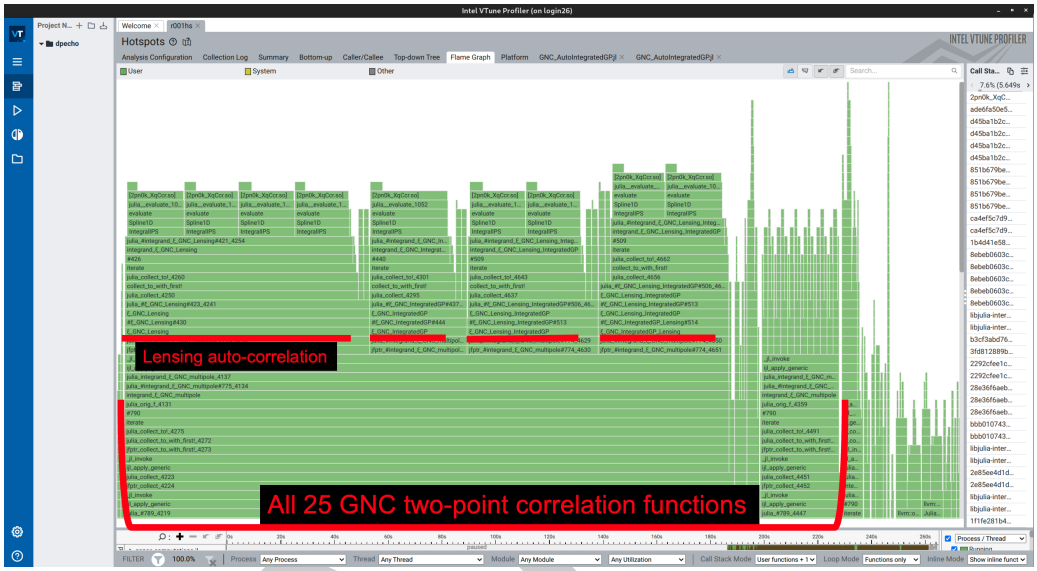


Figure 1: GaPSE.jl profiled with VTune

However, the evaluation at each input point can be done independently from the others. This provides a straightforward strategy for parallelization over the requested input vector of distances, without introducing dependencies between evaluations. For example, the following code snippet computes all the 25 GNC TPCF multipoles and their sum over a logarithmically spaced set of separations (and saves them to a file):

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[1]:
```
using GaPSE
const PATH_TO_GAPSE = normpath(joinpath(@__DIR__, ".."));  # GaPSE.jl location
println("Threads.nthreads() of this kernel : ", Threads.nthreads())
```

```
Threads.nthreads() of this kernel : 4
```

[2]:
```
params = GaPSE.CosmoParams(0.05, 0.20, pi / 2.0; # z_min, z_max, θ_max
    Ω_b = 0.0489, Ω_cdm = 0.251020,
    h_0 = 0.70, s_lim = 1e-2, z_spline_lim = 1000.0,
    b1 = 1.0, s_b1 = 0.0, f_evo1 = 0.0,
    b2 = nothing, s_b2 = nothing, f_evo2 = nothing,
);
COSMO = GaPSE.Cosmology(params,
    joinpath(PATH_TO_GAPSE, "data", "WideA_ZA_background.dat"),
    joinpath(PATH_TO_GAPSE, "data", "WideA_ZA_pk.dat"),
    joinpath(PATH_TO_GAPSE, "data", "F_REFERENCE_pi2.txt"),
    joinpath(PATH_TO_GAPSE, "data", "IntegrF_REFERENCE_pi2_z005020.txt")
);
vec_ss = 10 .^ range(0, log10(2 * COSMO.s_max), length=100);
GaPSE.print_map_sum_ξ_GNC_multipole(COSMO, "output.txt", vec_ss;
    L=0, use_windows=true, :obs => :noobsvel)
```

```
auto_newton, L=0: 100%|████████████████████| Time: 0:00:00
auto_lensing, L=0: 100%|████████████████████| Time: 0:00:05
auto_integratedgp, L=0: 100%|████████████████████| Time: 0:00:02
newton_lensing, L=0: 100%|████████████████████| Time: 0:00:00
lensing_newton, L=0: 100%|████████████████████| Time: 0:00:00
newton_integratedgp, L=0: 100%|████████████████████| Time: 0:00:00
integratedgp_newton, L=0: 100%|████████████████████| Time: 0:00:00
lensing_doppler, L=0: 100%|████████████████████| Time: 0:00:00
doppler_lensing, L=0: 100%|████████████████████| Time: 0:00:00
doppler_integratedgp, L=0: 100%|████████████████████| Time: 0:00:00
integratedgp_doppler, L=0: 100%|████████████████████| Time: 0:00:00
lensing_localgp, L=0: 100%|████████████████████| Time: 0:00:00
localgp_lensing, L=0: 100%|████████████████████| Time: 0:00:00
lensing_integratedgp, L=0: 100%|████████████████████| Time: 0:00:03
integratedgp_lensing, L=0: 100%|████████████████████| Time: 0:00:03
localgp_integratedgp, L=0: 100%|████████████████████| Time: 0:00:00
integratedgp_localgp, L=0: 100%|████████████████████| Time: 0:00:00

time needed for map_sum_ξ_GNC_multipole L=0[in s] = 24.08144
```

Figure 2: GaPSE.jl 25 GNC TPCFs computation, 4 threads

The evaluations over `vec_ss` are distributed across the available Julia threads. In the tested configuration, four threads provide a speedup of approximately $3.5\times$ relative to single-threaded execution.

The Power Spectrum multipoles are computed using FFTLog-based algorithms provided by `TwoFAST.jl` (Grasshorn Gebhardt & Jeong, 2018) and FFTLog.jl (Bonici, 2021). These methods evaluate the required transforms efficiently, making the Power Spectra calculation comparatively inexpensive relative to the nested integrations required for the TPCFs.

Numerous Jupyter notebooks are provided in the directories `ipynbs` and `theory` to demonstrate how to use the code and clarify important theoretical and computational aspects: among them, `ipynbs/TUTORIAL.ipynb` shows the basic workflow step-by-step, `ipynbs/Generic_Window.ipynb` and `ipynbs/eBOSS_Window.ipynb` how to use a user-supplied window function and an application to a real survey window, respectively, and `theory/threads_scaling.ipynb` the scaling of the parallel implementation. Finally, a Docker container with all the aforementioned features and files can be pulled from `foglienimatteo/gapse`.

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Research impact statement

GaPSE.jl has already supported published research. It was used to derive and analyze the large-scale limit of the observed galaxy power spectrum, including the potential degeneracy between relativistic contributions and local Primordial Non-Gaussianity (Foglieni et al., 2023). It was subsequently used to investigate the power spectrum of Luminosity Distance fluctuations in General Relativity and their relevance to peculiar-velocity surveys (Pantiri et al., 2024).

The study reported in Friedman-Shaw et al. (2025) motivated an extension of GaPSE.jl to handle multiple galaxy populations within the same Cosmology instance. This extension enables the computation of cross-correlations between two galaxy populations with different values of the parameters b_1, $s_{\rm b}$, $f_{\rm evo}$ (galaxy bias, magnification bias and evolution bias, respectively).

AI usage disclosure

No AI tools were used in the development of GaPSE.jl before and up to version v0.8.0. Beginning with version v0.10.0, we took advantage of Claude Opus 5.4 and ChatGPT Astra to investigate numerical issues and inconsistencies arising in specific corner cases, to improve the software documentation and assist in drafting and revising this manuscript. The authors reviewed and validated the AI-assisted contributions and take responsibility for the final software, documentation, and paper.

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